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# Pixel Value Reduction on OLED devices

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# Background

**2012** Optics and Optometry

Science University, Granada

**2019** Computer Science

Computer and Telecommunication Engineering
University, Granada

**2020** Computational Color and Spectral Imaging

Norwegian University of Science and Technology
Jean Monnet University, France
Uiversity Eastern of Finland

# Index

*"Energy fuels technology's ability to transform our lives and erase boundaries. If we ignore it, it can also stop us in our tracks."*

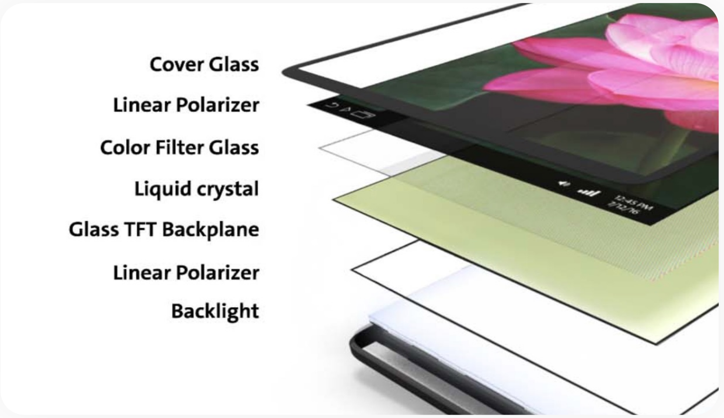
*- Interdigital*

## Green IT

Energy consumption of electronic devices has become an important consideration.
The need to reduce environmental impact arises.


## Display power consumption

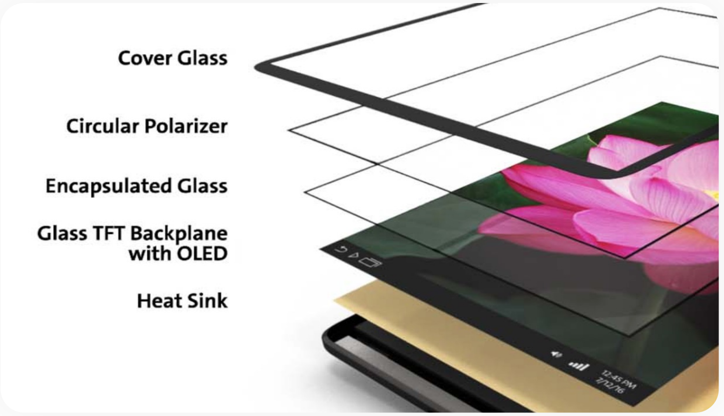
Displays on personal computers consume up to 38% of the total power while displays on mobile or handheld devices consume up to 50% of the total energy.




## LCD pannel

Backlight in LCDs consume almost 80% of the total energy (at 200nits brightness).

VS




## OLED technology

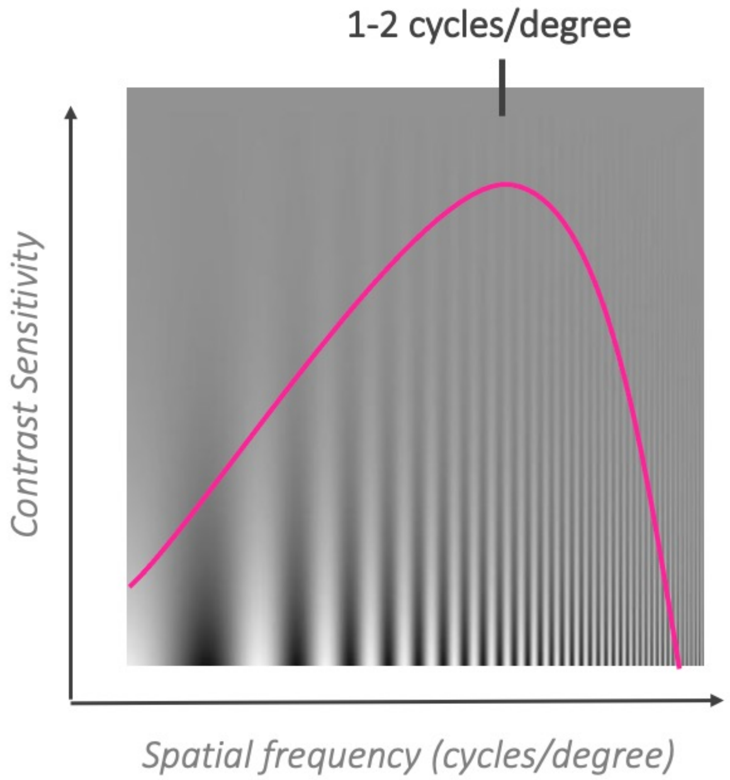
Does not require a constant light source.

Pixel power consumption proportional to its intensity level: Reduces energy pollution.

Better results in terms of contrast, brightness and good view.

# Contrast Sensitivity Function (CSF)

## Campbell-Robson chart




## Barten's Model

$$S(u) = \frac{1}{m_t} = \frac{M_{opt}(u)/k}{\sqrt{\frac{2}{T}\left(\frac{1}{X_0^2} + \frac{1}{X_{max}^2} + \frac{u^2}{N_{max}^2}\right)\left(\frac{1}{\eta p E} + \frac{\Phi_0}{1 - e^{-(u/u_0)^2}}\right)}}$$

$$M_{opt}(u) = e^{-2\pi^2\sigma^2 u^2}$$

$$\sigma = \sqrt{\sigma_0^2 + (C_{ab} d)^2}$$

$$d = 5 - 3\,tanh\left(0.4\,log\left(L\,X_0^2/40^2\right)\right)$$

$$E = \frac{\pi d^2}{4} L\left(1 - (d/9.7)^2 + (d/12.4)^4\right)$$

$k = 3.0$

$\sigma_0 = 0.5\ arc\,min$

$C_{ab} = 0.08\ arc\,min/mm$

$T = 0.1\ sec$

$X_{max} = 12°$

$N_{max} = 15\ cycles$

$\eta = 0.03$

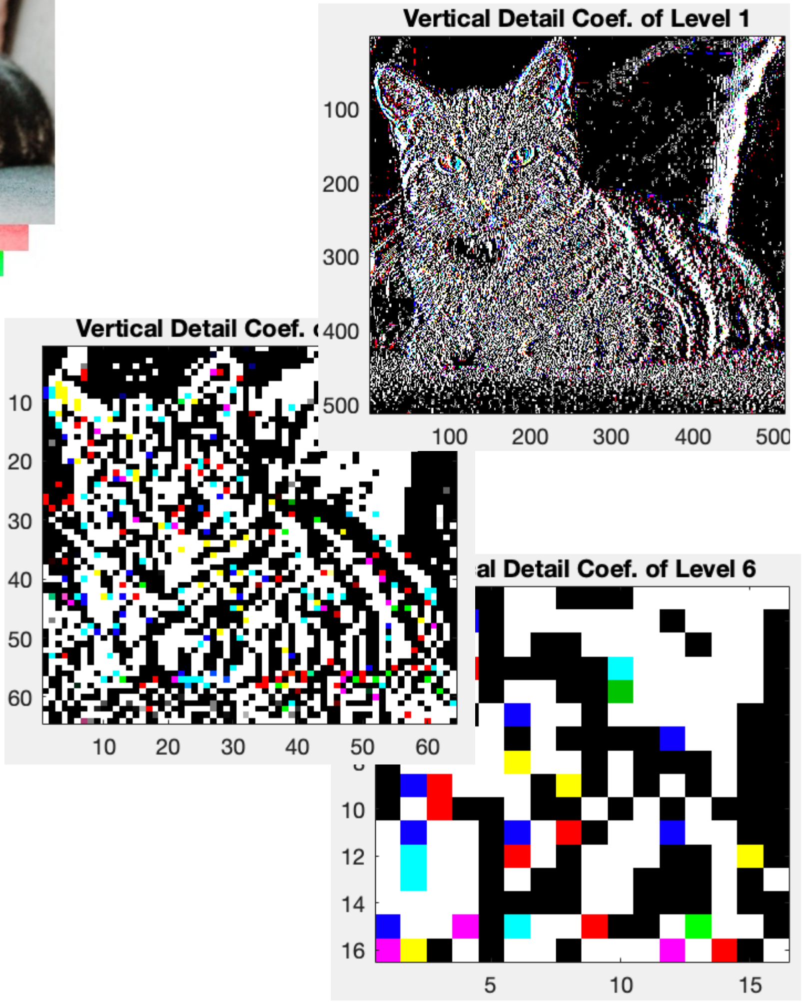
$\Phi_0 = 3 \times 10^{-8}\ sec\ deg^2$

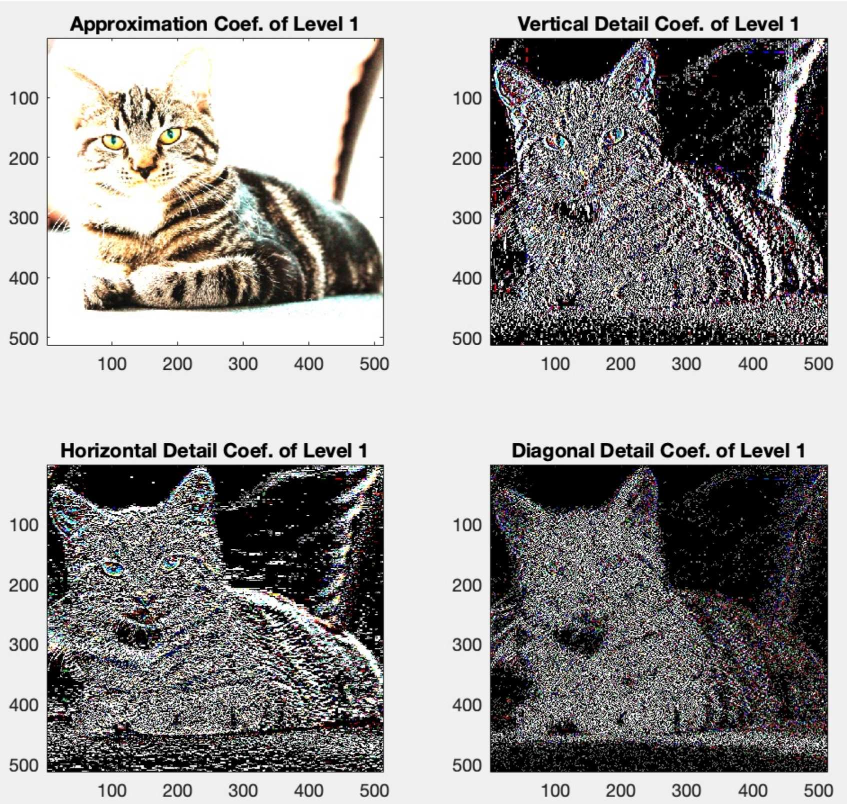
$u_0 = 7\ cycles/deg$

$p = 1.2 \times 10^6\ photons/sec/deg^2/Td$

# Discrete Wavelet Transform (DWT)

Analyses the image at **different resolutions.**




Allow us to extract how much there is of each **frequency** in each pixel location.

DWT

Luminance
$L$

Frequencies map
$u_i$

Barten's model
$S(L, u_i)$

$$u = \underset{u_i}{argmax}\, DWT(x, u_i) S(L, u_i)$$

Barten's model
$S(L, u)$

Human Contrast Sensitivity

Minimum detectable modulation

$$m(x) = \frac{1}{S(x)}$$

Michelson contrast

$$C_{michelson} = \frac{L_{high} - L_{low}}{L_{high} + L_{low}}$$

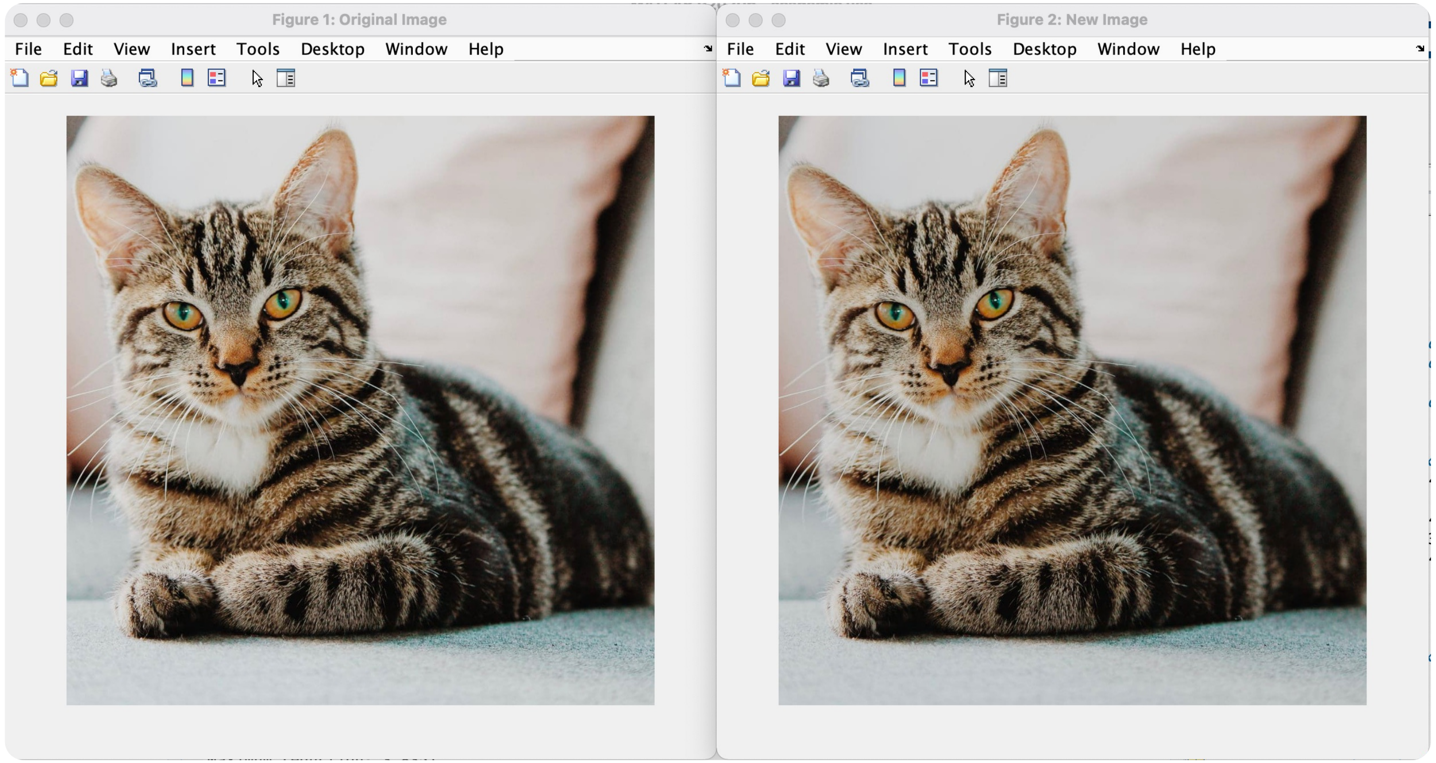
$$L_{low}(x) = L_{high}(x) \frac{1 - m(x)}{1 + m(x)}$$

Each pixel $x$ is changed independently

Difference less than 1 JND

4%

Image Luminance Reduction




# Results

The total luminance is decreased by 4%, while the output image imperceptibly different from the input image.

Successful outcome: relevant reduction in the total power consumption.

# Conclusions

Our method **reduces the total luminance** by approximately **4%.**

This result requires future **psychophysical experiments**.

Our approach could be **applied** at different stages in the imaging pipeline: prior encoding or in a user device.

The result is the production of less light and consequently the **reduction of energy consumption**.

# References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# Thank you for listening!

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